

Medvale Internal Medicine - Chapter 2J

Pulmonary Review, Integrated Cases, and Diagnostic Algorithms

Purpose. This closing pulmonary chapter synthesizes Chapters 2A-2I into a problem-based framework for internal medicine learners. It emphasizes diagnosis and management, ED/inpatient decision-making, numerical thresholds, and explicit concept relationships for future graph RAG extraction. The chapter is original Medvale writing; prior sources are used only as coverage signals and accuracy checks.

1. Why This Review Chapter Matters Clinically

Pulmonary medicine is often tested as a pattern-recognition problem: a patient is dyspneic, hypoxemic, coughing, wheezing, febrile, bleeding, or has an abnormal chest image. The correct answer usually depends on mapping the dominant physiologic problem to the safest next test or treatment. A patient with sudden pleuritic chest pain and syncope is not worked up the same way as a patient with chronic productive cough, a low FEV1/FVC ratio, and recurrent infections. A patient coughing up one tablespoon of blood can often undergo CT-based evaluation, while a patient drowning in blood requires immediate airway protection before the diagnostic workup is complete.

This chapter is designed as a capstone. The goal is not to repeat every disease description, but to connect the major pulmonary diseases through symptoms, mechanisms, tests, treatments, complications, and exam clues. Because the NBME-style internal medicine exam emphasizes diagnosis and management more than isolated mechanisms, the highest-yield approach is to ask four questions: Is the patient oxygenating? Is the patient ventilating? Is the process obstructive, restrictive, vascular, infectious, malignant, pleural, or sleep-related? What intervention must occur before the next diagnostic step?

Core clinical anchor: In pulmonary emergencies, the first branch point is stability. Unstable respiratory patients are managed by airway, breathing, circulation, oxygenation, ventilation, and hemodynamic support before definitive diagnostic elegance.

2. Master Symptom Algorithms

2.1 Approach to Acute Dyspnea

Acute dyspnea is a time-sensitive symptom because it may represent impending respiratory arrest, shock, pulmonary embolism, pneumothorax, asthma/COPD exacerbation, pneumonia, ARDS, pulmonary edema, or severe metabolic acidosis. The first task is to determine whether the patient is speaking in full sentences, protecting the airway, maintaining oxygen saturation, and generating adequate ventilation. Tachypnea is a warning sign, but respiratory rate alone can mislead: a patient with neuromuscular failure may look deceptively calm while carbon dioxide rises.

```
Acute dyspnea first-pass map
Dyspnea + hypotension -----> shock branch | PE,
tension pneumothorax, sepsis, cardiogenic edema | +-- wheeze + prolonged expiration ----->
asthma/COPD/bronchospasm branch | +-- fever + infiltrate ----->
pneumonia/aspiration branch | +-- pleuritic pain + risk factors ----->
PE/pneumothorax/pleuritis branch | +-- bilateral opacities + low P/F -----> ARDS vs
cardiogenic pulmonary edema branch | +-- chronic dry cough + crackles ----->
ILD/sarcoid/occupational branch
```

Immediate observation Most useful implication Common next action

----- Cannot speak, altered
 mental status, exhaustion, silent chest Impending ventilatory failure Prepare airway; support
 ventilation; do not wait for complete workup Hypotension + unilateral absent breath sounds Tension
 pneumothorax until proven otherwise Needle/finger decompression, then chest tube Hypotension +
 clear lungs + RV strain Massive PE or obstructive shock Bedside echo, anticoagulation, consider
 thrombolysis if PE likely Fever + focal crackles + consolidation Pneumonia likely Oxygen, cultures
 when indicated, empiric antibiotics, severity triage Diffuse wheeze + hyperinflation Obstructive
 exacerbation Bronchodilators, steroids, oxygen target, NIV if hypercapnic COPD

2.2 Approach to Chronic Dyspnea

Chronic dyspnea is best approached by tempo, exposures, spirometry, diffusion capacity, imaging, and cardiac overlap. Obstructive diseases produce airflow limitation, restrictive diseases produce reduced lung volumes, pulmonary vascular disease reduces perfusion and diffusion, and sleep-related disorders may present indirectly through fatigue, resistant hypertension, pulmonary hypertension, or chronic hypercapnia. A normal chest x-ray does not exclude asthma, PE, early ILD, pulmonary hypertension, anemia, or neuromuscular weakness.

Pattern Key clues Diagnostic direction Typical treatment direction -----

 Obstructive Wheeze, cough, prolonged expiration, FEV1/FVC below lower limit of normal or often
 <0.70 Spirometry with bronchodilator testing; consider DLCO Bronchodilators, inhaled steroids when
 indicated, trigger control, smoking cessation Restrictive/interstitial Dry cough, exertional
 dyspnea, inspiratory crackles, clubbing, low TLC, low DLCO High-resolution CT; exposure and
 autoimmune history Remove exposure, disease-specific therapy, antifibrotics for selected IPF,
 oxygen/transplant referral when advanced Vascular Pleuritic pain, syncope, RV strain, loud P2,
 unexplained low DLCO CTPA/VQ for PE; echo then right heart cath for pulmonary hypertension
 Anticoagulation for thromboembolism; pulmonary hypertension group-specific care
 Sleep/hypoventilation Snoring, witnessed apnea, daytime sleepiness, BMI ≥ 30 , awake PaCO₂ >45 mmHg
 in OHS Sleep testing, ABG or serum bicarbonate screening CPAP/BiPAP, weight loss, treat comorbid
 OSA/COPD/HF

3. Cough, Sputum, and Hemoptysis

3.1 Cough Pattern Recognition

Cough becomes high-yield when it is paired with duration, sputum quality, fever, wheeze, medication exposure, reflux symptoms, upper airway symptoms, hemoptysis, immune status, and imaging. Acute cough is most commonly infectious but may still represent PE, asthma, aspiration, or heart failure. Chronic cough lasting more than 8 weeks classically raises upper airway cough syndrome, asthma, GERD, nonasthmatic eosinophilic bronchitis, ACE inhibitor use, COPD, bronchiectasis, malignancy, and ILD.

Cough clue Strong associations Do not miss -----

----- Chronic productive cough
 Chronic bronchitis, bronchiectasis, cystic fibrosis, recurrent infection Bronchiectasis with
 hemoptysis or Pseudomonas risk Dry cough + crackles ILD, IPF, sarcoidosis, drug-induced
 pneumonitis Hypoxemia and pulmonary hypertension Cough after meals or altered consciousness
 Aspiration pneumonitis/pneumonia Lung abscess or anaerobic complication when necrosis/empyema
 present Cough + blood Bronchitis, bronchiectasis, pneumonia, cancer, TB, PE, vasculitis Massive
 hemoptysis requiring airway-first management Night cough + wheeze Asthma, GERD, postnasal drip
 Uncontrolled asthma or occupational trigger

3.2 Hemoptysis Algorithm

Hemoptysis is expectoration of blood originating below the vocal cords. The first step is to distinguish true hemoptysis from hematemesis, epistaxis, or oral bleeding. True hemoptysis is often bright red and frothy and may be accompanied by cough. Hematemesis is usually associated with nausea, coffee-ground material, or melena. In practice, the most important distinction is not the exact source but whether the airway is threatened.

Massive hemoptysis has variable definitions. Common numeric thresholds include approximately 100 to 600 mL over 24 hours, but physiology matters more than volume. Small volumes can kill if blood floods the airway, especially in patients with limited pulmonary reserve. The safest exam logic is: airway first, bleeding side down if known, large-bore IV access, correct coagulopathy, isolate the bleeding lung when possible, and involve interventional pulmonology, interventional radiology, and thoracic surgery early.

Hemoptysis pitfall: The usual cause of death in massive hemoptysis is asphyxiation, not exsanguination. A stable-looking oxygen saturation can deteriorate quickly when blood obstructs the tracheobronchial tree.

Hemoptysis category Likely causes Preferred evaluation Management emphasis

---- Scant or mild Bronchitis, pneumonia, airway irritation, mild bronchiectasis CXR; CT if risk factors, recurrent bleeding, abnormal imaging, age/smoking risk Treat cause; follow persistent or unexplained bleeding Moderate/recurrent Bronchiectasis, malignancy, TB, fungal disease, PE, vasculitis CT chest with contrast or CTA; bronchoscopy depending on stability and localization needs Localize source; treat infection/inflammation; evaluate malignancy Massive or airway-threatening Bronchiectasis, TB/cavitary infection, cancer, necrotizing pneumonia, vascular lesion Airway stabilization first; CTA and bronchoscopy often complementary Protect airway, bleeding lung down, reverse anticoagulation, bronchial artery embolization often definitive

4. Hypoxemia and Hypercapnia: The Numeric Framework

Hypoxemia means inadequate arterial oxygenation. Hypercapnia means inadequate alveolar ventilation relative to carbon dioxide production. The two frequently coexist but should not be treated as identical. Oxygen can improve hypoxemia without correcting ventilation; noninvasive ventilation or invasive ventilation may be required to remove CO₂. A patient with COPD can have acceptable oxygen saturation but worsening somnolence and acidosis from CO₂ retention.

Mechanism A-a gradient Typical response to oxygen Examples -----

----- Low inspired oxygen Normal Good High altitude Hypoventilation Normal Good for oxygen, but CO₂ may persist without ventilation Opioids, neuromuscular weakness, obesity hypoventilation V/Q mismatch Increased Usually improves COPD, asthma, pneumonia, PE Shunt Increased Poor or incomplete response ARDS, dense pneumonia, atelectasis, intracardiac shunt Diffusion limitation Increased, especially with exertion Often improves at rest ILD, emphysema, pulmonary vascular disease

ABG logic for pulmonary review 1. pH low? Acidic. pH high? Alkalemic. 2. PaCO₂ high? Respiratory acidosis component. 3. PaCO₂ low? Respiratory alkalosis component. 4. HCO₃ low? Metabolic acidosis component. 5. HCO₃ high? Metabolic alkalosis or chronic compensation. 6. Low PaO₂ or low SaO₂? Ask why oxygenation is failing. 7. COPD/OHS/neuromuscular patient with PaCO₂ rising + pH falling = acute ventilatory decompensation.

5. Integrated Test Interpretation

5.1 Pulmonary Function Tests

Spirometry is the central test for airflow obstruction. Obstruction is suggested when FEV1/FVC is reduced, often using <0.70 as a simple clinical threshold, though lower limit of normal is more precise. Bronchodilator reversibility is classically an increase in FEV1 or FVC of at least 12% and at least 200 mL, supporting asthma or a reversible component. Restriction requires lung volumes, especially reduced total lung capacity; spirometry alone can suggest but not prove restriction.

PFT pattern FEV1/FVC TLC DLCO Classic diagnoses -----

----- Asthma Low

during active obstruction; may normalize Normal or high if air trapping Normal or high Reversible obstruction; trigger-related symptoms COPD emphysema Low Normal/high Low Smoking, hyperinflation, dyspnea, poor recoil COPD chronic bronchitis Low Normal/high Often normal or mildly low Chronic productive cough Bronchiectasis Often low, mixed possible Variable Variable Recurrent infection, dilated bronchi on CT ILD/IPF Normal/high Low Low Dry cough, crackles, HRCT fibrosis Pulmonary hypertension Often normal Often normal Low Dyspnea, syncope, loud P2, RV strain Neuromuscular weakness Normal/high or nonspecific Low Normal if adjusted for volume Low maximal inspiratory/expiratory pressures

5.2 Chest Imaging Pattern Recognition

Chest imaging is powerful but must be interpreted with clinical context. A focal lobar opacity in a febrile patient suggests bacterial pneumonia, but a focal opacity in an older smoker with weight loss could represent post-obstructive pneumonia from lung cancer. Bilateral opacities with hypoxemia can represent ARDS, cardiogenic edema, diffuse alveolar hemorrhage, multifocal pneumonia, aspiration, or acute ILD exacerbation. Prior imaging is often the most valuable diagnostic test for nodules and chronic abnormalities.

Image pattern Likely clinical associations Key next questions

----- Lobar consolidation

Pneumococcal pneumonia, aspiration, obstructing tumor Fever? aspiration risk? cancer risk? follow-up resolution? Interstitial reticulation/honeycombing IPF/UIP, connective tissue disease ILD, asbestosis Age, exposures, autoimmune symptoms, HRCT pattern Hyperinflation COPD, asthma exacerbation, air trapping Smoking, alpha-1 risk, wheeze, PFTs Pleural line without vascular markings Pneumothorax Tension signs? secondary lung disease? trauma/procedure? Blunted costophrenic angle Pleural effusion Transudate vs exudate; tap if new, unilateral, febrile, large, or atypical Solitary pulmonary nodule Granuloma, hamartoma, primary lung cancer, metastasis Size, growth, calcification, smoking risk, prior CT

6. High-Yield Management Ladders

6.1 Oxygenation Versus Ventilation

Oxygenation is primarily improved by increasing FiO₂, recruiting alveoli with PEEP, treating shunt or V/Q mismatch, and resolving the underlying cause. Ventilation is primarily improved by increasing minute ventilation, reducing dead space, supporting respiratory muscles, and correcting airway obstruction. Confusing these two domains leads to common errors, such as escalating oxygen flow in a patient whose main problem is CO₂ narcosis.

Problem Useful tools Numerical anchors Common mistake -----

----- Hypoxemia Nasal cannula, face mask, high-flow nasal cannula, CPAP/PEEP, intubation ARDS severity uses PaO₂/FiO₂: mild 200-300, moderate 100-200, severe <100 Not recognizing shunt physiology when oxygen response is poor Hypercapnia Bronchodilators, NIV, invasive ventilation, reverse sedatives/opioids when appropriate Acute ventilatory failure: rising PaCO₂ with acidemia, often pH <7.35 Giving oxygen alone and ignoring worsening ventilation COPD oxygenation Controlled oxygen, bronchodilators, steroids, antibiotics when indicated, NIV Often target SpO₂ 88-92% in known CO₂ retainers Over-oxygenation without ABG monitoring in severe hypercapnia risk ARDS ventilation Lung-protective ventilation, PEEP, proning for severe ARDS, treat cause Tidal volume about 6 mL/kg predicted body weight; plateau pressure generally <30 cm H₂O Using actual body weight or excessive tidal volume

6.2 Noninvasive Ventilation Versus Intubation

Noninvasive ventilation is most useful in acute COPD exacerbation with hypercapnic respiratory acidosis and in cardiogenic pulmonary edema. It may help selected immunocompromised or post-extubation patients, but it is risky when airway protection is poor, secretions are heavy, shock is present, mental status is severely impaired, or immediate invasive ventilation is needed. A trial of NIV must have a reassessment point; failure to improve pH, respiratory effort, or mental status should trigger escalation.

Scenario NIV usually helpful? Reasoning -----

----- COPD exacerbation + pH 7.25-7.35 + elevated PaCO₂ Yes Reduces work of breathing and improves ventilation Cardiogenic pulmonary edema Yes Positive pressure recruits alveoli and reduces preload/afterload ARDS with severe hypoxemia Often limited High failure risk; delayed intubation can worsen outcomes Massive hemoptysis No Airway contamination and aspiration risk; intubation/localization may be required Vomiting, coma, inability to protect airway No Aspiration risk and poor airway control

7. Disease Comparison Tables

7.1 Pneumonia Versus PE Versus CHF Versus COPD

Feature Pneumonia Pulmonary embolism Cardiogenic pulmonary edema COPD/asthma exacerbation

----- Onset Hours-days Sudden or subacute Acute/subacute Hours-days Symptoms Fever, cough, sputum, pleuritic pain Pleuritic pain, dyspnea, syncope, hemoptysis Orthopnea, edema, frothy sputum Wheeze, chest tightness, cough Exam Crackles, bronchial breath sounds Tachycardia, hypoxemia; lungs may be clear Crackles, S3, JVD, edema Wheeze, prolonged expiration, accessory muscles Testing CXR infiltrate, cultures if severe Wells/PERC/D-dimer/CTPA/VQ BNP, echo, CXR vascular congestion Spirometry history; ABG if severe Immediate treatment Antibiotics + oxygen + fluids if septic Anticoagulation; thrombolysis if massive and appropriate Diuresis, nitrates, CPAP/BiPAP if needed Bronchodilators, steroids, oxygen target; NIV for COPD acidosis

7.2 Pleural Effusion Rapid Interpretation

Effusion clue Likely diagnosis Key fluid patterns Action

----- Bilateral, known CHF, improves
with diuresis Transudative CHF effusion Low protein/LDH by Light's criteria Diuresis; tap if
atypical, unilateral, febrile, or nonresolving Pneumonia + pleural fluid pH <7.20 or glucose low
Complicated parapneumonic effusion/empyema Neutrophilic, high LDH, possible pus/positive culture
Drainage plus antibiotics Malignancy symptoms or recurrent unilateral effusion Malignant pleural
effusion Exudate; cytology may diagnose Thoracentesis, cytology, symptom palliation Very high
amylase Pancreatitis or esophageal rupture Exudate with elevated amylase Treat source; esophageal
rupture is emergency Milky fluid Chylothorax High triglycerides Evaluate trauma, malignancy,
thoracic duct injury

7.3 Lung Cancer and Pulmonary Nodule Review

Lung cancer questions often combine risk factors, histology, paraneoplastic syndromes, and local invasion. Adenocarcinoma is common and often peripheral; squamous cell carcinoma is classically central and associated with hypercalcemia from PTHrP; small cell carcinoma is central, aggressive, neuroendocrine, and associated with SIADH, ectopic ACTH, and Lambert-Eaton syndrome. Pulmonary nodules are risk-stratified by size, growth, morphology, calcification pattern, patient age, smoking history, prior imaging, and immune status.

Clinical clue Diagnosis to consider Mechanistic link

----- Shoulder pain + Horner
syndrome + hand weakness Pancoast tumor Apical tumor invades brachial plexus/sympathetic chain
Facial swelling + venous distention SVC syndrome Central tumor compresses superior vena cava
Hyponatremia in smoker with lung mass Small cell carcinoma with SIADH Neuroendocrine ectopic ADH
activity Hypercalcemia in central cavitary mass Squamous cell carcinoma PTHrP-mediated
hypercalcemia Stable calcified nodule over years Benign granuloma/hamartoma more likely Chronicity
and benign calcification reduce malignancy probability

8. Integrated Case Practice

Case 1: Acute dyspnea after surgery

A 67-year-old woman is 5 days post hip surgery and develops sudden dyspnea, pleuritic chest pain, tachycardia, and oxygen saturation of 88% on room air. Her lungs are clear. The key concept is that clear lungs do not exclude severe pulmonary disease. Postoperative immobility increases venous thromboembolism risk, and sudden pleuritic symptoms suggest PE. If she is stable, CT pulmonary angiography is usually the direct diagnostic test. If renal failure or contrast allergy is a major limitation and the chest x-ray is interpretable, V/Q scanning may be used. If unstable, bedside echo evidence of RV strain can support emergent management while definitive imaging may be unsafe.

Answer logic: PE is a vascular obstruction problem causing V/Q mismatch, increased dead space, RV pressure overload, and sometimes obstructive shock. Anticoagulation is treatment unless contraindicated; thrombolysis is reserved for massive PE with hemodynamic instability or selected high-risk deterioration.

Case 2: Wheezing patient who is getting quiet

A 24-year-old with asthma arrives with severe wheezing, then becomes drowsy with less audible wheeze. This is not improvement. A silent chest can signal minimal air movement. Worsening hypercapnia in asthma reflects respiratory muscle fatigue and severe airflow obstruction. Initial therapy includes inhaled short-acting beta agonist, ipratropium for severe exacerbation, systemic corticosteroids, oxygen, and magnesium in severe cases. If mental status worsens or ventilation fails, intubation is required by an experienced team because dynamic hyperinflation and auto-PEEP can cause hypotension after intubation.

Case 3: Fever, aspiration, and anaerobic confusion

A nursing-home patient with dysphagia coughs after eating and develops fever and a right lower lobe infiltrate 2 days later. This is aspiration pneumonia, not just chemical pneumonitis. Aspiration pneumonitis is an acute chemical injury that often follows a witnessed macroaspiration event and may improve within 24-48 hours without antibiotics if bacterial infection does not develop. Aspiration pneumonia is bacterial infection after aspiration and is treated like pneumonia with attention to oral flora and severity. Routine extended anaerobic coverage is not required for every aspiration event, but it becomes more important when lung abscess, necrotizing infection, empyema, severe periodontal disease, or classic anaerobic complication is present.

Case 4: Hemoptysis in bronchiectasis

A 58-year-old with chronic productive cough and CT-proven bronchiectasis presents with repeated large-volume hemoptysis. The airway is the immediate threat. Place the bleeding lung down if known, protect the airway, reverse coagulopathy, and use CT angiography and bronchoscopy as the patient's stability allows. Bronchial artery embolization is often the key definitive intervention because most severe hemoptysis arises from the high-pressure bronchial arterial circulation rather than the low-pressure pulmonary arterial circuit.

Case 5: Bilateral opacities and low PaO₂/FiO₂

A septic patient develops acute hypoxemia, bilateral opacities, and a PaO₂/FiO₂ ratio of 90 despite positive pressure ventilation. Cardiac function is not the primary explanation. This fits severe ARDS. Management includes treating sepsis, lung-protective ventilation using predicted body weight, limiting plateau pressure, using adequate PEEP, considering prone positioning for severe ARDS, conservative fluid strategy after shock resolves, and monitoring for ventilator-induced lung injury.

9. High-Yield Numeric Anchors

Number Meaning Clinical use -----

----- FEV₁/FVC <0.70 Common
simple threshold for obstruction COPD/asthma/bronchiectasis airflow limitation FEV₁ or FVC increase ≥12% and ≥200 mL Classic bronchodilator response Supports reversible airway disease PaCO₂ >45 mmHg while awake with BMI ≥30 Key OHS anchor after excluding other causes Obesity hypoventilation syndrome SpO₂ 88-92% Common target range for COPD patients at risk of CO₂ retention Controlled oxygen strategy Light's criteria positive Pleural exudate Pleural protein ratio >0.5, LDH ratio >0.6, or LDH >2/3 upper limit normal Pleural pH <7.20 Complicated parapneumonic effusion Usually needs drainage PaO₂/FiO₂ 200-300, 100-200, <100 Mild, moderate, severe ARDS categories ARDS severity and management escalation Tidal volume about 6 mL/kg PBW Lung-protective ventilation anchor ARDS and many ventilated patients Plateau pressure <30 cm H₂O Ventilator lung protection target Reduce barotrauma/volutrauma risk Hemoptysis 100-600 mL/24h Common massive hemoptysis range Airway threat matters more than exact volume Minimum 5 days Common CAP antibiotic duration when clinically stable Requires stability and afebrile/improving status Age 50-80, 20 pack-years, current or quit <15 years USPSTF lung cancer screening criteria Annual low-dose CT in eligible adults

10. Common Pitfalls

- Calling every wheeze asthma. CHF, PE, foreign body, anaphylaxis, vocal cord dysfunction, and COPD can wheeze. - Ignoring ventilation because oxygen saturation improved. Oxygenation and CO₂ removal are different physiologic tasks. - Using D-dimer in high-probability PE as a rule-out test. D-dimer is most useful when pretest probability is low or sometimes intermediate. - Delaying decompression for suspected tension pneumothorax. Tension physiology is a clinical diagnosis in an unstable patient. - Assuming bilateral opacities are always CHF. ARDS, diffuse pneumonia, aspiration, hemorrhage, and ILD exacerbation can mimic edema. - Overusing anaerobic coverage for aspiration. Reserve broader anaerobic logic for abscess, empyema, necrosis, or severe periodontal/anaerobic features. - Underreacting to hemoptysis. The airway can be lost before the blood pressure changes. - Calling a pleural effusion transudative without context. Diuretics can concentrate pleural fluid and complicate interpretation. - Missing pulmonary hypertension in chronic lung disease. Progressive dyspnea, syncope, loud P₂, edema, and low DLCO should trigger consideration. - Not checking old imaging for a pulmonary nodule. Stability over time can drastically change risk.

11. Graph RAG Extraction Map

The following concept map is included to make node and edge extraction easier. It highlights how pulmonary symptoms, mechanisms, diagnostic tests, treatments, and complications connect.

Node Useful edges High-yield attributes -----

----- Dyspnea caused_by asthma, COPD, pneumonia, PE, CHF, pneumothorax, ARDS, ILD, anemia, acidosis acute vs chronic; stable vs unstable; oxygenation vs ventilation Hypoxemia mechanism V/Q mismatch, shunt, diffusion limitation, hypoventilation, low FiO₂ A-a gradient; oxygen response; PaO₂/FiO₂ in ARDS Hypercapnia caused_by COPD, OHS, neuromuscular weakness, sedatives, severe asthma PaCO₂, pH, bicarbonate compensation Hemoptysis caused_by bronchiectasis, TB, cancer, pneumonia, PE, vasculitis; treated_by airway protection, embolization massive vs nonmassive; asphyxiation risk Aspiration subtype pneumonitis vs pneumonia; complication lung abscess, empyema witnessed event; dependent lobes; anaerobic coverage conditions Pleural effusion classified_by Light's criteria; caused_by CHF, malignancy, pneumonia, PE, cirrhosis pH, glucose, LDH, cytology, drainage indications Pulmonary embolism arises_from DVT; causes V/Q mismatch, RV strain, obstructive shock Wells, PERC, D-dimer, CTPA, anticoagulation, thrombolysis ARDS caused_by sepsis, pneumonia, aspiration, pancreatitis, trauma; treated_by lung-protective ventilation timing, bilateral opacities, noncardiogenic edema, P/F ratio Obstructive lung disease includes asthma, COPD, bronchiectasis; measured_by FEV₁/FVC wheeze, air trapping, bronchodilator response, DLCO pattern Interstitial lung disease causes restriction, low DLCO, pulmonary hypertension HRCT pattern, exposure, autoimmune disease, crackles

12. Closing Synthesis

The pulmonary section can be reduced to a few high-yield axes. First, decide whether the problem is acute, chronic, or acute-on-chronic. Second, separate oxygenation failure from ventilation failure. Third, determine whether the chest is obstructed, restricted, infected, vascularly occluded, filled with fluid, compressed by pleural air/fluid, infiltrated by tumor, or under-ventilated during sleep. Fourth, match severity to site of care. An outpatient with stable chronic cough requires a different approach than an inpatient with rising CO₂, hypotension, or airway-threatening hemoptysis.

From a learning standpoint, the most important pulmonary skill is not memorizing isolated lists, but building links. Bronchiectasis links chronic sputum to recurrent infection, *Pseudomonas*, obstructive PFTs, and hemoptysis. PE links DVT risk to pleuritic pain, clear lungs, RV strain, and anticoagulation. ARDS links sepsis or aspiration to bilateral opacities, refractory hypoxemia, low PaO₂/FiO₂, and lung-protective ventilation. Sarcoidosis links granulomas to bilateral hilar lymphadenopathy, uveitis, erythema nodosum, hypercalcemia, and steroid-responsive organ disease. These links are the core of clinical reasoning and the ideal structure for a graph RAG knowledge base.

Selected References Used for Medical Accuracy Checks

1. American Thoracic Society / European Respiratory Society guideline materials on noninvasive ventilation for acute respiratory failure. 2. American College of Radiology Appropriateness Criteria: Hemoptysis. 3. American Thoracic Society / Infectious Diseases Society of America guideline for diagnosis and treatment of adults with community-acquired pneumonia. 4. Guideline anchors from prior Medvale pulmonary chapters: GOLD COPD strategy, GINA asthma strategy, Fleischner pulmonary nodule guidance, USPSTF lung cancer screening recommendation, ATS/ERS/JRS/ALAT ILD/IPF and hypersensitivity pneumonitis guidance, ATS sarcoidosis and obesity hypoventilation guidance. 5. Uploaded pulmonary reference packet used as broad coverage signal only; wording and figures were not reproduced.

Medvale Internal Medicine - Chapter 2J

Medvale note: This study resource does not replace local protocols, specialist consultation, or clinical judgment.